

# Assignment 2: End-to-End ML Versioning

Evidence-based final report - Git, DVC, Google Drive and Fashion-MNIST ANN

|              |                               |             |                                         |
|--------------|-------------------------------|-------------|-----------------------------------------|
| Student      | Zainab Fatima                 | Repository  | github.com/Aenonic/fashion-ann-pipeline |
| Dataset      | Fashion-MNIST (70,000 images) | Final model | Dense 256, 10 epochs                    |
| Final result | Accuracy 87.81% / loss 0.3474 | Audit date  | 4 October 2026                          |

## Part A - Git Fundamentals and Advanced Commands

The repository has an unsquashed history. The reflog independently preserves the destructive-history exercises (soft/hard reset) and the completed rebase; the merge commit preserves the Part E conflict resolution.

| Requirement      | Verified evidence                                                                                                                                                 | Meaning                                                                            |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| A1-A2            | Initial commit d21ef1b on main; dev branch created; 9 incremental Part B-D commits before v2.                                                                     | Branch isolation and incremental history are present.                              |
| A3 log variants  | <b>--oneline --graph --all</b> : topology; <b>--stat -3</b> : recent file counts; <b>-p -1</b> : exact patch; <b>main..dev</b> : commits reachable only from dev. | Each exposes structure or change detail omitted by plain log.                      |
| A4 diff variants | <b>git diff</b> : unstaged; <b>--staged</b> : index; <b>main..dev</b> : tip-to-tip; <b>main...dev</b> : merge-base-to-dev.                                        | Two-dot includes both tips; three-dot isolates the dev-side work since divergence. |
| A5 stash         | Workflow documented: stash push -u, stash list, branch switch, stash pop.                                                                                         | The report records the required sequence; no stash entry remains after pop.        |
| A6 rebase        | Reflog: rebase started onto hotfix 570787f and finished with dev at 6e78009.                                                                                      | Dev was replayed on top of updated main.                                           |
| A7 reset         | Reflog records soft reset and hard reset on scratch-reset-demo; branch ends at Throwaway commit 1.                                                                | Soft retained staged changes; hard discarded commit and worktree changes.          |
| A8 mv/rm         | Commits 7e2b603 and 6e78009 retain preparation and reorganization events.                                                                                         | Tracked move/removal history is retained.                                          |

### Representative command evidence

```
$ git log --oneline --graph --all
* 932d0d5 (HEAD -> main, origin/main) Docs: update report / Drive remote
* f3fe902 Docs: add PDF report and generator
* d76cdaf E5: pipeline reproducible from merged state
* dcc7948 E5: resolve code and DVC data conflicts
| * c7c77ec (teammate-sim) E1: normalization [-1,1]
* | 49a91eb E2: main Z-score normalization
|/
* b81c629 (tag: v2, dev) D5: dense_units=256, accuracy=0.8781
* faac07e (tag: v1) D3: dense_units=128, accuracy=0.8740

$ git reflog --all | evidence summary
6e78009 rebase (finish) onto 570787f
fa5cd83 -> d5399cf reset HEAD~1 (soft demonstration)
fa9b02b -> d5399cf reset HEAD~1 (hard demonstration)
```

## Part B - Modular TensorFlow ANN Pipeline

| Script            | Inputs                         | Outputs / verified behavior                                                              |
|-------------------|--------------------------------|------------------------------------------------------------------------------------------|
| src/prepare.py    | tf.keras Fashion-MNIST         | Four raw .npy arrays; 60k training and 10k test examples.                                |
| src/preprocess.py | raw arrays, params.yaml        | Float32 [0,1] scaling; stratified 20% validation split; six processed arrays.            |
| src/train.py      | processed arrays, train params | Flatten -> Dense(ReLU) -> Dropout -> Dense(10, Softmax); Adam; model.h5 and history.csv. |
| src/evaluate.py   | model + processed test set     | Loss/accuracy in metrics.json and labeled confusion matrix PNG.                          |

## Part D - Reproducible DVC pipeline

### params.yaml

```
preprocess:
  test_size: 0.2
  seed: 42

train:
  dense_units: 256
  dropout_rate: 0.2
  learning_rate: 0.001
  epochs: 10
  batch_size: 64
```

### dvc.yaml

```
stages:
  prepare:
    cmd: python3 src/prepare.py
    deps:
      - src/prepare.py
    outs:
      - data/raw

  preprocess:
    cmd: python3 src/preprocess.py
    deps:
      - data/raw
      - src/preprocess.py
    params:
      - preprocess.test_size
      - preprocess.seed
    outs:
      - data/processed

  train:
    cmd: python3 src/train.py
    deps:
      - data/processed
      - src/train.py
    params:
      - train.batch_size
      - train.dense_units
      - train.dropout_rate
      - train.epochs
      - train.learning_rate
    outs:
      - models/history.csv
      - models/model.h5

  evaluate:
    cmd: python3 src/evaluate.py
    deps:
      - data/processed
      - models/model.h5
      - src/evaluate.py
    outs:
      - reports/confusion_matrix.png
    metrics:
      - metrics.json:
          cache: false
```

### D3/D4 execution and caching

At v1 (tag faac07e), a clean reproduction created dvc.lock with dense\_units=128 and test accuracy 0.8740. For v2 (tag b81c629), only **train** and downstream **evaluate** were invalidated by dense\_units 128 -> 256; prepare and preprocess remained cached because their dependencies and parameters did not change.

| Version | Dense | Dropout | LR    | Epochs | Loss    | Accuracy | Target |
|---------|-------|---------|-------|--------|---------|----------|--------|
| v1      | 128   | 0.20    | 0.001 | 10     | 0.3411  | 87.40%   | Met    |
| v2      | 256   | 0.20    | 0.001 | 10     | 0.3474  | 87.81%   | Met    |
| Delta   | +128  | -       | -     | -      | +0.0063 | +0.41 pp | Met    |

### Current lock-file evidence

```
prepare      raw hash 585a7ccf...dir
preprocess   processed hash 673bfbf0...dir (test_size=0.2, seed=42)
train        model hash bf4f29dc... (dense_units=256)
evaluate     metrics hash 94391a7f...; confusion matrix hash 3469b359...
metrics.json test_accuracy=0.87809998, test_loss=0.34735397
```

## Part C - DVC Setup and Google Drive Remote

DVC is initialized and the default remote is configured as **gdrive\_storage** with folder ID **1A4889YAEgPAiFNWejJkhAuGMXrLowjNe**. A dedicated Google Cloud desktop OAuth client was authorized without committing credentials to Git. The final **dvc push** uploaded 14 files, **dvc status -c** reported the cache and remote in sync, and the supplied Drive folder visibly contains the DVC **files** directory.

| Check                | Status               | Evidence / required action                                                                |
|----------------------|----------------------|-------------------------------------------------------------------------------------------|
| DVC initialized      | Verified             | .dvc exists and dvc version identifies the repository.                                    |
| Default Drive remote | Verified             | .dvc/config points to the supplied Drive folder ID.                                       |
| Credentials excluded | Verified             | .gitignore excludes credentials.json, client_secrets.json, token.json and token patterns. |
| dvc push             | Verified             | Completed successfully: 14 files pushed.                                                  |
| Drive contents       | Verified             | Google Drive shows the DVC files directory; remote status is in sync.                     |
| Instructor sharing   | User action required | Share the folder with the instructor account if it is not already shared.                 |

## Part E - Simulated Collaboration and Conflict Resolution

The conflict is genuine: teammate-sim used zero-centered  $[-1,1]$  scaling while main independently used Z-score normalization. Both branches regenerated data/processed, yielding different DVC hashes. Merge commit dcc7948 reconciled the code to canonical  $[0,1]$  scaling and selected regenerated processed hash 673bfb0...dir; d76cdf then recorded the reproducible final state.

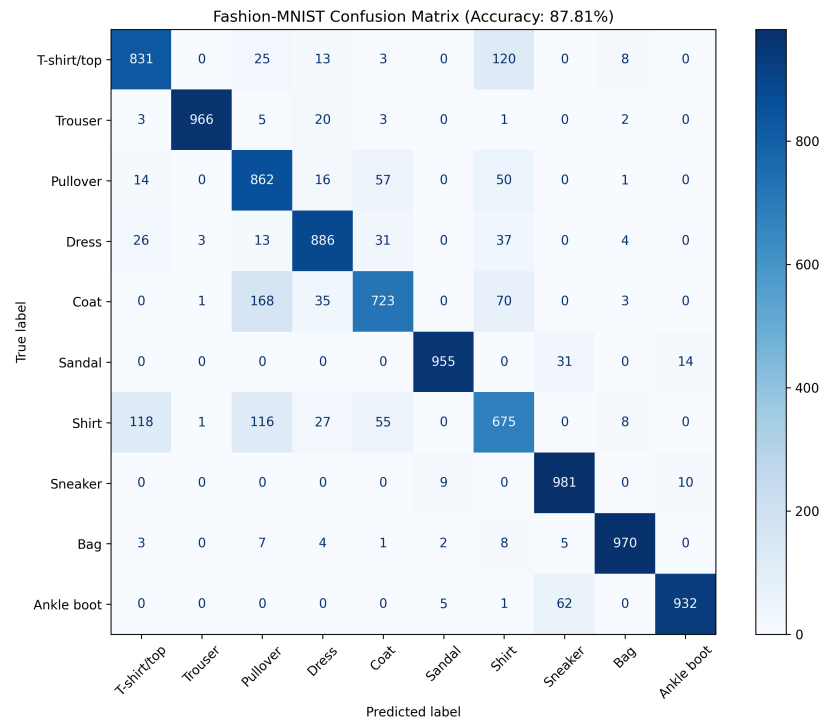
```
[CONFLICT DEMONSTRATION: Part E3]
$ git merge teammate-sim
Auto-merging data/processed.dvc
CONFLICT (content): Merge conflict in data/processed.dvc
Auto-merging src/preprocess.py
CONFLICT (content): Merge conflict in src/preprocess.py
Automatic merge failed; fix conflicts and then commit the result.

--- data/processed.dvc ---
<<<<<< HEAD (main)
- md5: dec45f84294f6511cbd8d5c0e9997e7f.dir
=====
+ md5: 31645b3bbc540c83585eb31cb97a35f4.dir
>>>>>> teammate-sim

--- src/preprocess.py ---
<<<<<< HEAD (main: Z-score normalization)
mean = np.mean(train_images)
std = np.std(train_images) + 1e-7
train_images = (train_images.astype('float32') - mean) / std
test_images = (test_images.astype('float32') - mean) / std
===== (teammate-sim: Zero-centered [-1, 1] normalization)
train_images = (train_images.astype('float32') - 128.0) / 128.0
test_images = (test_images.astype('float32') - 128.0) / 128.0
>>>>>> Teammate-sim
```

Conflict evidence: Git markers are visible in both `src/preprocess.py` and `data/processed.dvc`. The final merge commit has two parents (49a91eb and c7c77ec), and the working source no longer contains conflict markers.

## Final Artifacts and Compliance Audit



The confusion matrix corresponds to the v2 test accuracy of 87.81%. The weakest class separation is Shirt versus visually similar upper-body categories; this is expected for a fully connected ANN and does not affect the assignment threshold.

### Submission checklist

| Deliverable                            | Status               | Notes                                                                         |
|----------------------------------------|----------------------|-------------------------------------------------------------------------------|
| GitHub repository + unsquashed history | Verified             | origin/main is synchronized; v1/v2 and collaboration branches exist remotely. |
| Four standalone pipeline scripts       | Verified             | All required scripts and model architecture are present.                      |
| params.yaml + dvc.yaml + dvc.lock      | Verified             | Four stages are wired; final lock is committed.                               |
| Accuracy >= 85%                        | Verified             | v2 accuracy 87.81%.                                                           |
| Google Drive DVC objects               | Verified             | 14 files pushed; cache and gdrive_storage are in sync.                        |
| Instructor Drive access                | User action required | Share the Drive folder with the instructor account.                           |
| Report accuracy                        | Corrected            | Repository, metrics, upload, and Drive evidence were cross-checked.           |

**Final conclusion.** The Git history, ANN implementation, DVC pipeline definition, lock file, metrics, tags, conflict-resolution history, Google Drive upload, and remote synchronization meet the technical requirements. The only remaining submission-side check is to ensure the supplied Drive folder is shared with the instructor account.